Performance Analysis of 3D Mesh Simplification Algorithms

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Problem & Motivation

The Problem

High-polygon 3D models create performance bottlenecks in real-time graphics applications

Research Question

Do performance and geometric fidelity of *decimation* algorithms differ significantly between clean CAD models vs. noisy user-generated models?

599,821 Faces

Decimation Algorithms Compared

Quadric Error Metrics

QEM

26,538 faces

- Feature-preserving edge collapse algorithm
- Topology-aware: optimizes vertex placement to minimize geometric error
- Maintains surface details and sharp features

Original

53,078 faces

Vertex Clustering Decimation

Clustering

19,965 faces

- Spatial grid-based clustering approach
- Topology-agnostic: merges vertices by proximity, not connectivity
- Extremely fast but less precise feature preservation

Methodology & Evaluation

Datasets

- ModelNet40 CAD (15 models) - Clean CAD
- Thingi10K (15 models) - Organic/Scanned

Two-Factor Design

- Algorithm (2 levels: QEM, Clustering)
- Mesh Type (2 levels: Clean CAD, Organic/Scanned)

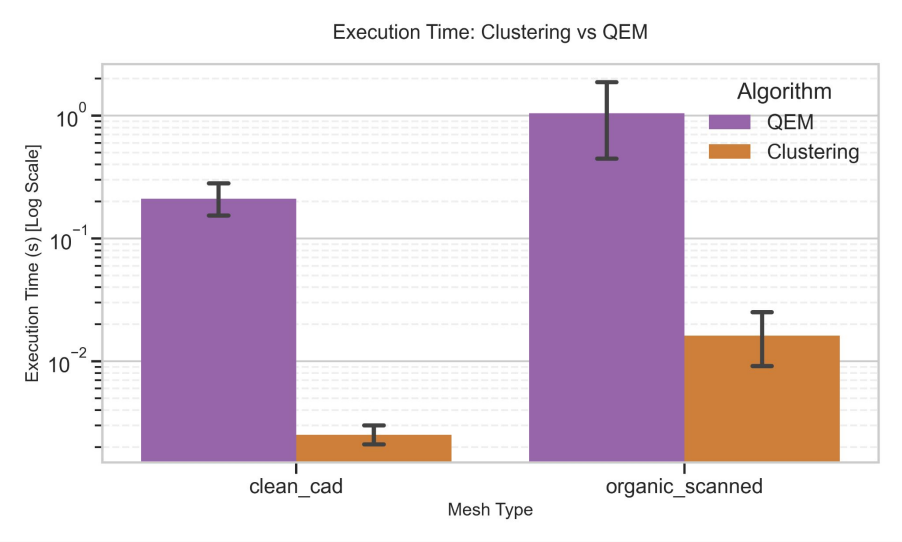
Metrics

- Execution Time (wall-clock, 5 repetitions, warm-up runs)
- Two-Sided Hausdorff Distance (geometric fidelity)

Statistical Analysis (α=0.05)

- Two-Way ANOVA for main effects and interactions
- Welch's T-tests for targeted comparisons
- 95% Confidence Intervals for all metrics

Execution Time Results



CLUSTERING: *Consistently fast*

- Clean CAD: 0.0025s (2.5 milliseconds)
- Organic: 0.0161s (16.1 milliseconds)

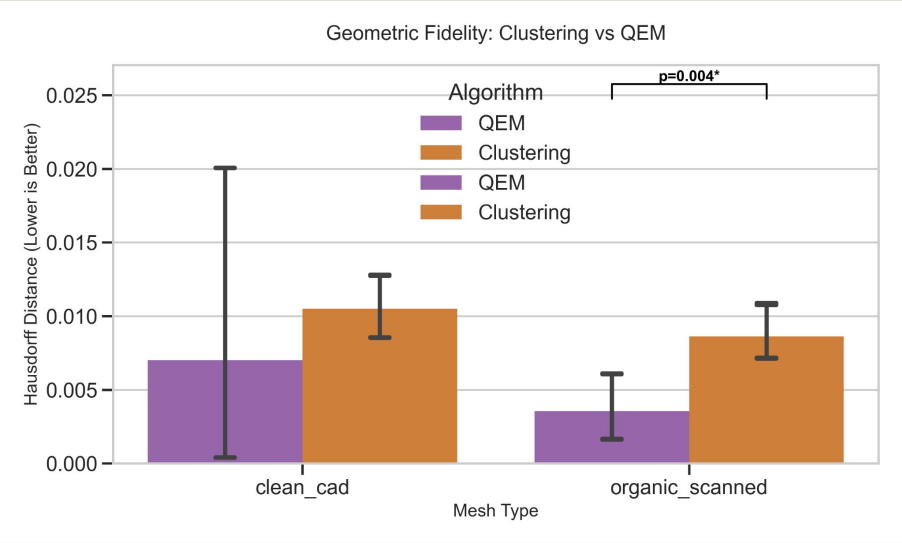
QEM: *Dramatic performance gap*

- Clean CAD: 0.2103s (210.3 milliseconds)
- Organic: 1.0466s (over 1 second)

STATISTICAL SIGNIFICANCE:

- Algorithm effect: $F(1,56)=11.57$, **$p=0.0012$** (S)
- Interaction effect: $F(1,56)=5.11$, **$p=0.0277$** (S)
- QEM is ~5x slower on Organic than on CAD

Geometric Fidelity Results



QEM preserves surface details better on complex organic geometry!

Mean Hausdorff Distance (Error)

CLUSTERING

- Clean CAD: 0.0105 ± 0.0043
- Organic: 0.0086 ± 0.0041

QEM

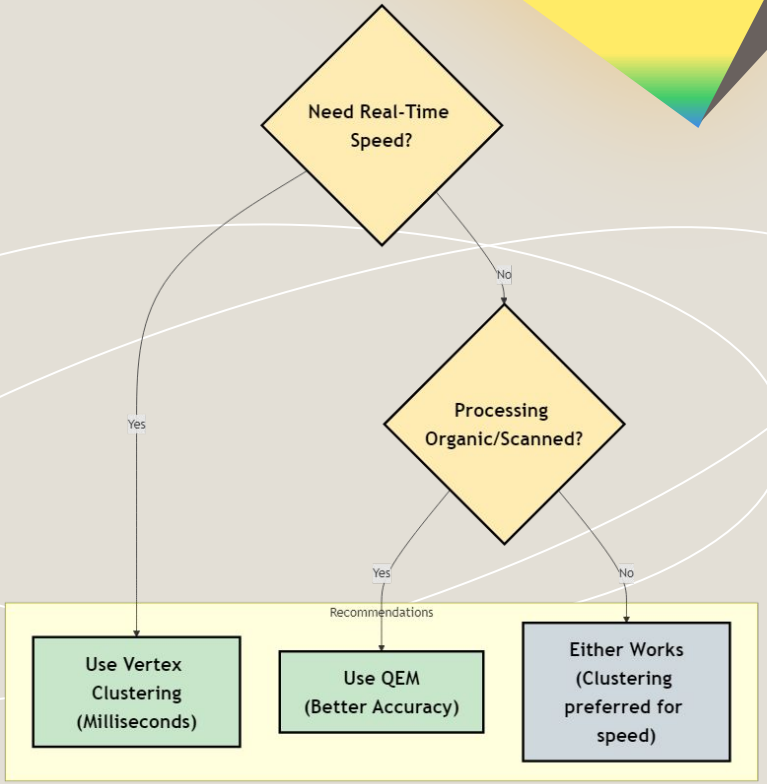
- Clean CAD: 0.0070 ± 0.0251
- Organic: 0.0036 ± 0.0047

Two-Way ANOVA Results

- Algorithm: F(1,56)=1.60, **p=0.211** (NS)
- Mesh Type: F(1,56)=0.62, **p=0.435** (NS)
- Interaction: F(1,56)=0.05, **p=0.818** (NS)

Conclusions & Recommendations

Key Findings

- Speed-quality trade-off confirmed
- Significant interaction effect
- Context-dependent performance

Limitations: Non-normality in data distribution, high variance in QEM processing times for complex meshes (n=15 per group)